Lab # 07

Stacks Implementation

Spring 2026

Data Structures And Algorithms Lab

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Task 01 Stack via Linked List Arrays:

```cpp
#include <iostream>
using namespace std;
#define SIZE 100
struct Stack
{
    int arr[SIZE];
    int top;
};

void createStack(Stack *s)
{
    s->top = -1;
}
void push(Stack *s, int value)
{
    // if (s->top >= 0 || s->top <= SIZE)
// invalid condition
    if (s->top <= SIZE - 1)
    {
        s->top++;
        s->arr[s->top] = value;
    }
    else
        cout << "Push function ----Stack
overflow" << endl;
}
void pop(Stack *s)
{
    // s->arr[s->top] = 0;
    // s->top--;

    if (s->top == -1)
    {
        cout << "pop fucntion ----Stack
underflow\n";
    }
```

```cpp
    s->arr[s->top--];
}
int top(Stack *s)
{
    if (s->top == -1)
    {
        cout << "top function ----Stack is
empty! No top element." << endl;
        return -1;
    }
    return s->arr[s->top];
}
int size(Stack *s)
{
    return s->top + 1;
}
void empty(Stack *s)
{
    s->top = -1;
}
int main()
{
    Stack s;
    createStack(&s);
    push(&s, 10);
    push(&s, 20);
    push(&s, 30);
    cout << "Top element: " << top(&s) <<
endl;
    empty(&s); // clears everything
    if (size(&s) == 0)
    {
        cout << "Stack is now empty." <<
endl;
    }
    return 0;
}
```

Output:

```
PS D:\4th Semester\DSA\lab_07_stacks_via_LLArrays> .\a.exe
Top element: 30
Stack is now empty.
```

Task 02 Stack implementation via Link List through Pointers:

```cpp
#include <iostream>
using namespace std;

struct Node
{
    int data;
    Node *next;
};

struct Stack
{
    Node *top;
    int count;
};

void createStack(Stack *s)
{
    s->top = nullptr;
    s->count = 0;
}

void push(Stack *s, int value)
{
    Node *newNode = new Node;
    newNode->data = value;
    newNode->next = s->top;
    s->top = newNode;
    s->count++;
}

void pop(Stack *s)
{
    if (s->top == nullptr)
    {
        cout << "Stack Underflow! Cannot pop" << endl;
        return;
    }
    Node *temp = s->top;
    s->top = s->top->next;
    delete temp;
    s->count--;
}
```

```cpp
int top(Stack *s)
{
    if (s->top == nullptr)
    {
        cout << "Stack is empty! No top element." << endl;
        return -1;
    }
    return s->top->data;
}

int size(Stack *s)
{
    return s->count;
}

void empty(Stack *s)
{
    while (s->top != nullptr)
    {
        pop(s); // repeatedly delete nodes
    }
}

int main()
{
    Stack s;
    createStack(&s);

    push(&s, 100);
    push(&s, 200);
    push(&s, 300);
    cout << "Top element: " << top(&s) << endl;
    empty(&s); // clears the stack fully
    if (size(&s) == 0)
    {
        cout << "Stack is now empty." << endl;
    }
    return 0;
}
```

Output:

```
PS D:\4th Semester\DSA\lab_07_stacks> .\a.exe
Top element: 300
Stack is now empty.
```

RUBRICS (LABS/ PROJECTS):

Criteria & Point Assigned	**Outstanding 4**	**Acceptable 3**	**Considerable 2**	**Below Expectations 1**	**Score**
Valuing related to work, including responsibility, and professionalism.	Demonstrates **excellent** responsibility and professionalism in task managment. Shows **strong commitment to** quality work, meeting the deadline, and comprehensive, well-organized **effective documentation** following all guidelines with exemplary attention to detail.	Demonstrates **some** responsibility and professionalism. Displays **moderate commitment to** quality work, meets the deadline and shows **sufficient thoroughness** in documentation, adhering to guidelines, with some room for enhancement.	Demonstrates **limited effort** in responsibility and professionalism. Misses the deadline with **notable inconsistency** in quality work and shows **insufficient** thoroughness or clarity in documentation, necessitating improvements.	Demonstrates a **lack** of responsibility and professionalism. Misses the deadline and **lacks** quality work. Documentation is incomplete, disorganized, or unclear.	
Ability to apply techniques, methods, and procedures effectively.	Executes lab tasks **with exceptional skill** to design/develop solution and **shows proficiency** in lab techniques and procedures.	Executes lab tasks **with adequate skill** to design/develop solution and **shows proficiency** in lab techniques and procedures.	Executes lab tasks **with minimal skill** to design/develop solution and **shows modeate proficiency** in lab techniques and procedures.	Executes lab tasks **with limited skill** to design/develop solution and **lacks proficiency** in lab techniques and procedures.	

Demonstrate understanding and relevant Knowledge.	Demonstrates **exceptional** understanding of concepts with **clear explanation** of investigation and analysis of the objectives and outcomes.	Demonstrates a **good** understanding of concepts with **some explanations** of investigation and analysis that align with the lab's objectives and outcomes.	Demonstrates a **basic** understanding of key concepts but **lacks clarity or depth in explaining** the investigation and analysis, with some inconsistencies.	Demonstrates **limite d** understanding of concepts with **unclear explanations** of investigation and analysis that do not align with the lab’s objectives and outcomes.	